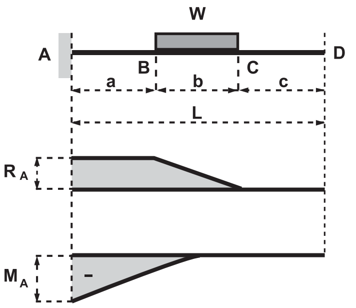
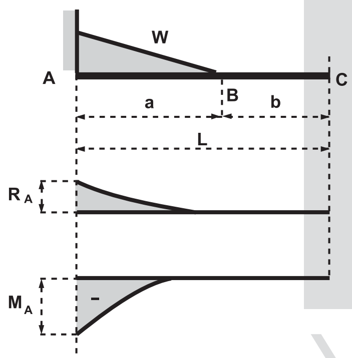
I-BEAMS

Cantilever	Uniform load on part of span		Special case: Uniform loan on full span	
	Span	$= L$	Span	$= L = b$
			a	$= c = 0$
	Uniform load	$= W$	Uniform load	$= W$
	R_A	$= W$	R_A	$= W$
	M_A	$= -W\left(a+\frac{b}{2}\right)$	M_A	$= -\frac{WL}{2}$
	$i_C = i_D =$	$\frac{W}{6EI}(3a^2+3ab+b^2)$	i_D	$= \frac{WL^2}{6EI}$
			δ_D	$= \frac{WL^3}{8EI}$



$$\delta_D = \frac{WL}{24EI}[8a^3+18a^2b+12ab^2+3b^3+4c(3a^2+3ab+b^2)]$$

Cantilever	Triangular load on part of span		Special case: Triangular loan on full span	
	Span	$= L$	Span	$= L = a$
			b	$= 0$
	Triangular load	$= W$	Triangular load	$= W$
	R_A	$= W$	R_A	$= W$
	M_A	$= -\frac{Wa}{3}$	M_A	$= -\frac{WL}{3}$
	δ_C	$= \frac{Wa^2}{15EI}\left(L+\frac{b}{4}\right)$	δ_C	$= \frac{WL^3}{15EI}$
	$i_B = i_C$	$= \frac{Wa^2}{12EI}$	i_C	$= \frac{WL^2}{12EI}$



Cantilever	Point load at any position		Special case: Point load at free end	
	Span	$= L$	Span	$= L = a$
			b	$= 0$
	Point load	$= W$	Point load	$= W$
	R_A	$= W$	R_A	$= W$
	M_A	$= -Wa$	M_A	$= -WL$
	δ_C	$= \frac{Wa^2}{3EI}\left(L+\frac{b}{2}\right)$	δ_C	$= \frac{WL^3}{3EI}$
	$i_B = i_C$	$= \frac{Wa^2}{2EI}$	i_C	$= \frac{WL^2}{2EI}$